Marker-Guided Indoor Motion Planning for Micro Aerial Vehicles with Monocular RGB Vision

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INTRODUCTION

Motion planning for Micro Aerial Vehicles (MAVs) in indoor environments refers to the set of algorithms and control strategies that enable drones to safely navigate enclosed or partially known spaces. Its applications include industrial inspection in factories and warehouses, search and rescue in confined areas, indoor logistics, as well as academic competitions and research testbeds. The main challenges in this context arise from the limited sensor payload imposed by weight constraints, the absence of GPS signals indoors, the sensitivity of vision-based approaches to illumination changes, and the need to ensure real-time performance despite restricted onboard computing resources.

PROPOSED APPROACH

The objective is to present a motion planning scheme for aerial indoors navigation in semi-unknown environments. The focus is on using computer vision to detect fiducial markers and visual patterns located at floor level using only a RGB monocular camera. Thus allowing a MAV to maintain and follow an arbitrary reference with respect to the camera frame. The approach demonstrates that low sensor requirements can achieve reliable indoor navigation, making this approach accessible for resource-constrained MAV platforms.

- The proposed system leverages a ROS2-PX4 communication framework to enable seamless integration between high-level navigation and low-level flight control through standardized message interfaces.
- OpenCV-based detection algorithms for two complementary visual markers are implemented: blue floor lines for continuous path-following and ArUco markers for precise waypoint positioning and orientation control.
- A comprehensive finite state machine architecture coordinates between 25+ distinct navigation states, managing transitions between line-following, marker alignment, task execution, and special maneuvers like window traversal.
- A PID controller with adaptive gain tuning processes HSV color space filtered images to maintain the MAV centered on blue floor lines. For ArUco marker alignment, a custom pose controller is implemented using quaternion-based orientation feedback and 3D position estimation to achieve precise positioning at configurable distances relative to detected markers.
- ROS2 nodes design that can be independently enabled/disabled, allowing dynamic switching between position control, velocity control, and specialized task controllers.

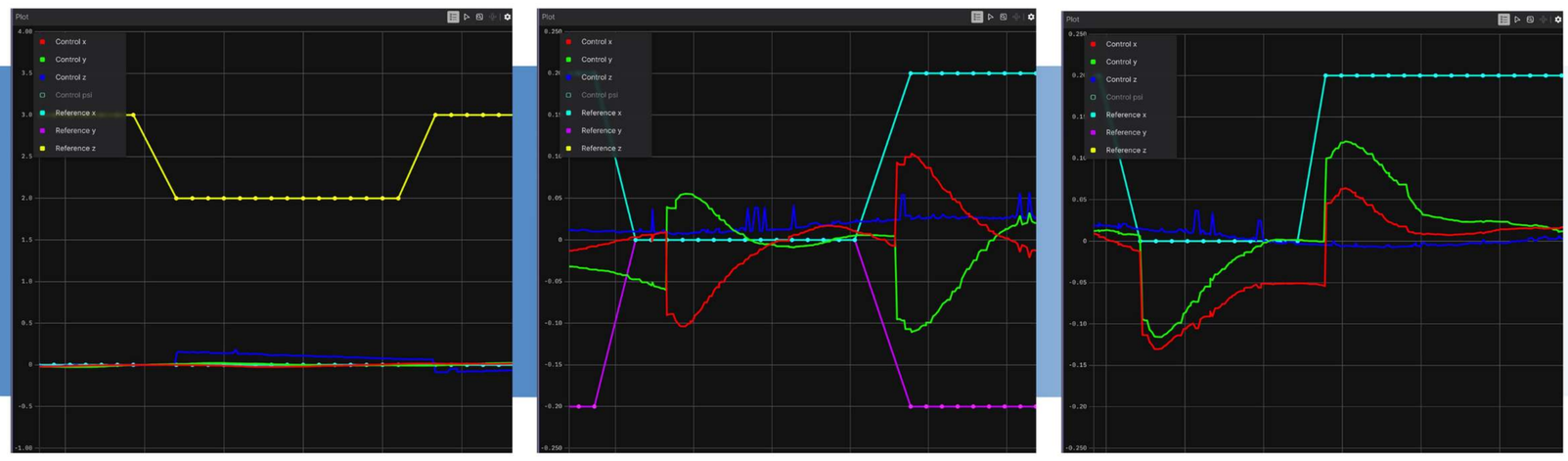


Fig. 1. MAV velocity control actions given set points.

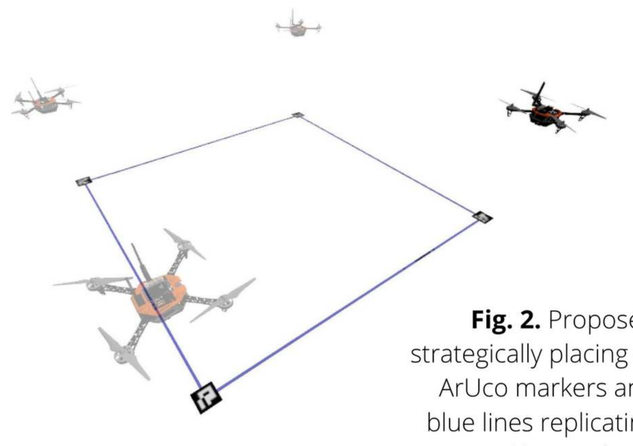


Fig. 2. Proposed strategically placing of ArUco markers and blue lines replicating competition and real life scenarios.

RESULTS

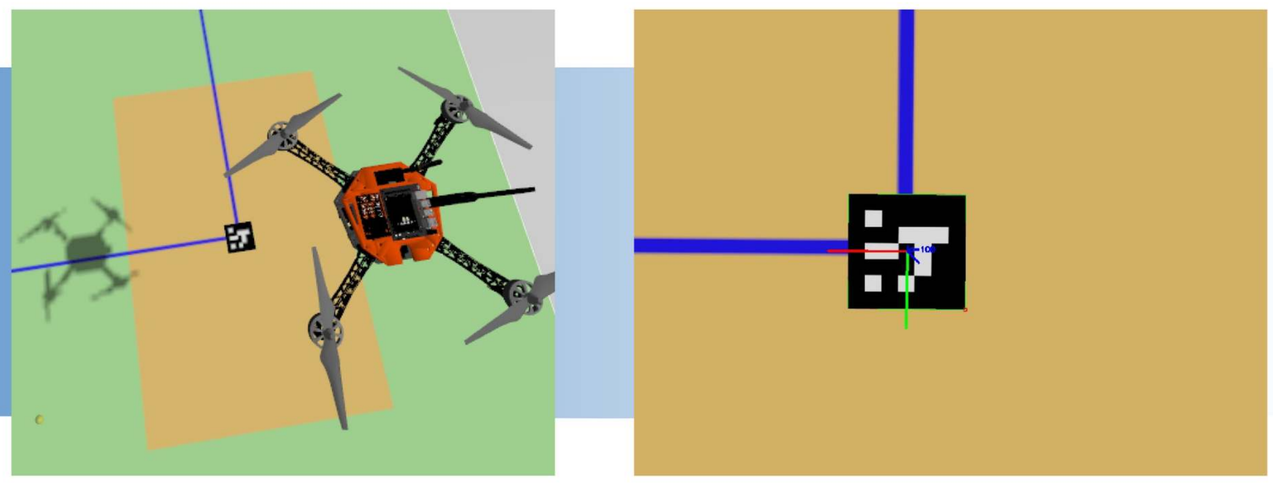


Fig. 3. Gazebo simulation for proposed scheme validation.



Fig. 4. Torneo Mexicano de Robótica, Autonomous Drones Competition 2025.

The proposed work successfully demonstrated complete mission execution in a Gazebo environment, where accurate MAV dynamics and sensor modeling effectively replicated competition scenarios. The motion planning scheme proved its effectiveness by securing a 5th overall place at the Torneo Mexicano de Robótica 2025 in the Autonomous Drones category, validating its real world applicability under competitive conditions.

CONCLUSION

This approach demonstrates that combining visual markers with a modular control architecture enables indoor navigation with reduced computational and sensor requirements. However, key limitations lie in the dependence on pre-installed visual references and sensitivity to illumination conditions. As a next step, integration of adaptive vision algorithms is considered to improve robustness under variable lighting, alongside the use of marker-independent localization techniques (e.g., VIO, SLAM) to reduce infrastructure dependency.



References

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